

Internal proof audit for the universal contact-geometric bound

Repository source: proof/PROOF_AUDIT.md

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Audit date: September 19, 2026 **Status:** internal project audit; not independent subject-matter review

Audit conclusion

No contradiction was found between the mathematical manuscript, the exact finite certificate, the machine-readable theorem fields, the independent binary64 and base-R audits, or the adversarial mutation tests.

The exact package reconstructs the proposed coefficient

$$\frac{26221074253\pi}{200000000000} = 0.4118796712122863775432315385981549189 \dots$$

and verifies that both split terminal rows 03B02A and 03B02B strictly clear the theorem coefficient. The unchanged near-Jung branch is therefore the universal bottleneck within the certified gap-free circumradius partition.

This conclusion is limited in four ways:

- the audit was performed inside the same project that produced the proof;
- much of the construction and exposition was AI-assisted;
- it is not a systematic novelty review;
- it does not establish peer review, external reproduction, sharpness, or Meissner extremality.

The public status remains:

Proposed exact universal certificate — standalone internal replay.

1. Audit layers

Layer	Object checked	Method	Result
Exact finite proof	Lower rows, fan, multipliers, determinant transfer, coordinate widths, caps, disjointness, assembly	Python integers and <code>fractions.Fraction</code>	Pass
Derived authority	Assembled certificate JSON and exact audit tables	Byte-identical reconstruction	Pass
Numerical implementation	Full theorem assembly	Independent NumPy binary64 path	Pass
Independent language	Archived decimals, cap allocations, and assembly	Base R only	Pass on named author's machine
Regression	Reconstruction and public-path invariants	Ten Python unit tests	Pass
Adversarial	Seven deliberately corrupted proof objects	Production semantic and byte-identity gates	All rejected
Manuscript	Coefficient, split, partition, and main constants	Generated TeX and reconciliation checker	Pass
Documentation	Required files, claim language, links, paired PDFs	Release-document and pair checkers	Pass in this checkpoint

The exact and numerical layers are intentionally separate. Agreement between them is evidence against implementation error, but only the exact path carries the finite inequality claims.

2. Geometric reduction audit

A1. Constant width and scaling

The proof normalizes the width to one and restores the factor d^3 by dilation. The support-function identity, diameter-one consequence, and Blaschke volume– surface relation agree with the standard sources listed in [SOURCE_LEDGER.md](#).

Status: source-checked and manuscript-checked.

A2. Minimum circumsphere and contact balance

The origin is placed at the center of a minimum circumscribed ball. Contact balance and Carathéodory's theorem reduce the relevant near-Jung regime to four contact directions. The circumradius range terminates at Jung's value $\sqrt{6}/4$.

Audit focus:

- no symmetry assumption is introduced;
- the four-contact conclusion is used only in the stated radius regime;
- the lower and near-Jung intervals meet exactly at $R_* = 305695601/500000000$.

Status: manuscript derivation and exact partition checked.

A3. Opposite-point and radial inequalities

The opposite-point identity converts constant width into pointwise support and ball bounds. These feed a divergence inequality and a concave tangent majorant for the radial function.

Audit focus:

- normal orientation and inequality direction;
- admissible parameter ranges;
- use of regularization when the boundary is not smooth;
- no floating-point quantity is used to certify a row.

Status: source comparison, manuscript check, and exact-row replay passed.

3. Lower-branch certificate audit

The lower interval is covered by thirteen rows. Four refined cells replace the formerly weak 03A1 interval. Their endpoints match exactly and the full table covers

$$\left[\frac{1}{2}, \frac{305695601}{500000000} \right]$$

without a gap or overlap affecting the argument.

The exact verifier checks each rational step certificate, cap bound, disjointness condition, and coefficient comparison. It then proves that 03B02B is the weakest lower-branch row, while the near-Jung branch sets the theorem coefficient.

High-risk items checked:

- exact endpoint adjacency;
- angular and radial grid denominators;
- sign conditions before squaring inequalities;
- downward rounding directions;
- coefficient-of- π versus decimal-volume columns;
- unique-bottleneck logic.

The first adversarial test changes a lower-row endpoint and is rejected by the semantic verifier after bypassing only the immutable-file fingerprint.

Status: exact replay, independent numerical audit, and mutation rejection passed.

4. Near-Jung fan audit

The final fan contains 98,306 rays, 196,608 triangular faces, 294,912 edges, and 24,833 distinguished-edge stabilizer orbits. The exact minimum fan margin is positive and exceeds the package threshold.

High-risk items checked:

- fan-array fingerprints and integer dtypes;
- edge-stabilizer orbit census;
- inherited and revised radial numerators;
- exact S-procedure multipliers;
- active-bound classification;
- strict positivity of every accepted margin.

Two adversarial tests alter a fan radial numerator and an S-procedure multiplier. Each mutation is rejected by the underlying exact semantic checks, not merely by a checksum mismatch.

Status: exact reconstruction and mutation rejection passed.

5. Determinant and coordinate-width audit

The determinant transfer uses the total edge deficiency S and the exact bound

$$D(e) \geq 1 - \frac{S}{2} - \frac{S^2}{2}.$$

After relabeling a maximum-deficiency edge as 01, the new analytic handoff uses

$$H_{00} \leq 1 + \frac{S}{2}, \quad H_{11}, H_{22} \leq 1 + \frac{S}{6},$$

and positive definiteness to infer reciprocal coordinate-width lower bounds.

Audit focus:

- the relabeling is exhaustive rather than a symmetry assumption;
- the dominant-edge inequality applies to the selected maximum edge;
- the two nondominant axes use the weaker but valid $S/6$ diagonal estimate;

- the inverse-diagonal inequality $(H^{-1})_{jj} \geq 1/H_{jj}$ has the correct direction;
- the coordinate factors are combined with the same core geometry used in the certificate.

The exact JSON, binary64 audit, paper-generated constants, and human-readable appendix agree on all three coordinate-width lower bounds.

Status: algebraically reviewed and exact-code checked. This is a priority for independent mathematical review because it is the main analytic step that separates the proposed coefficient from the predecessor certificate.

6. Cap allocation and disjointness audit

Three positive/negative coordinate-axis pairs are optimized separately. Their exact lower bounds sum to

0.0001439609020552581838284512468198759823427280084172919

The six open cap interiors are pairwise disjoint by exact transformed-ellipsoid inequalities. The smallest stored strict gap is positive.

High-risk items checked:

- cap bracket orientation;
- section-hull vertex order and exact polygon area;
- cap-pair summation;
- strict versus non-strict disjointness statements;
- no cap contribution is counted twice;
- the cap total is added to, rather than substituted for, the forced core.

Four mutation tests alter a section-hull vertex, cap bracket, disjointness threshold, and final theorem digit. The full reconstruction gate rejects every case.

Status: exact replay and mutation rejection passed.

7. Universal assembly audit

The exact near-Jung branch gives

0.4118796712148324317327979810531508741628745212952960018

and exceeds the upper enclosure of the lower-branch bottleneck by

0.0000021074345301574846190550703110031602996750911392740

Therefore both split terminal rows clear the displayed coefficient across the lower branch, while the near-Jung certificate supplies the universal bottleneck across the two closed intervals. The proof restores d^3 by scaling.

Audit focus:

- strict handoff direction;
- use of an upper enclosure for the bottleneck in the comparison;
- exact endpoint coverage;
- no interpolation across an uncertified radius interval;
- exact numerator and denominator in every public location.

Status: byte-identical certificate reconstruction, exact arithmetic, binary64 audit, base-R audit, and paper reconciliation agree.

8. Portability and release repairs

The following nonmathematical defects or risks were found and addressed during public-package construction:

- private parent-package imports were flattened into a standalone public tree;
- private branch names, commit hashes, and unrelated unreleased research material were removed;
- a seven-case mutation suite was added to ensure that proof-object corruption is detected;
- theorem-critical manuscript constants were moved to generated TeX includes;
- an early paper table with visual overlap was redesigned before packaging;

- the first generated exact-constants PDF used tables too wide for the page; the generator was changed to vertical row records, large exact fractions are identified by certificate field, and every lower-row volume decimal is now recomputed uniformly from its exact coefficient and the certificate's exact rational lower bound for π ;
- PDF byte identity across platforms was explicitly separated from mathematical validity;
- the local Mac's missing Poppler tools were documented, with full font/text preflight delegated to the hosted Ubuntu document job;
- Markdown/PDF one-to-one pairing and checksums were automated.

No repair silently changed the theorem. A future change to the authoritative certificate requires a new version and explicit coefficient comparison.

9. Literature and precedence audit

The source ledger was checked against official publisher, arXiv, and public repository records on September 26, 2026. That targeted search did not locate a later public universal coefficient exceeding the proposed one. This is not a systematic novelty review and should not be reported as one.

The targeted search was repeated on September 26, 2026, immediately before release. Independent experts should still be asked specifically about:

- unpublished or recently accepted universal lower bounds;
- alternate contact-geometric certificates;
- changes to the final bibliographic status of recent papers;
- precedence for the dominant-edge coordinate-width lemma.

10. Priority questions for external reviewers

1. Does the minimum-circumsphere reduction cover every nonsmooth constant-width body after the stated regularization?
2. Are all sign and monotonicity hypotheses in the lower-row propagation certificates stated strongly enough in the manuscript?
3. Is the dominant-edge coordinate-width lemma valid under every allowed contact labeling, including ties?
4. Does the determinant transfer use exactly the Gram normalization declared in the paper and JSON?
5. Do the transformed section hulls and cap-disjointness inequalities exclude all cross-axis overlaps?
6. Is the exact handoff comparison made against the correct enclosure of the lower bottleneck?
7. Are any recent universal bounds or relevant predecessor lemmas missing from the source ledger?

11. Current scientific boundary

The audit supports releasing the package, after the remaining metadata and hosted-CI gates, as an **AI-assisted, non-peer-reviewed computer-assisted research draft seeking independent mathematical verification**.

It does not support describing the result as independently certified, accepted, sharp, or a solution of the Meissner conjecture.